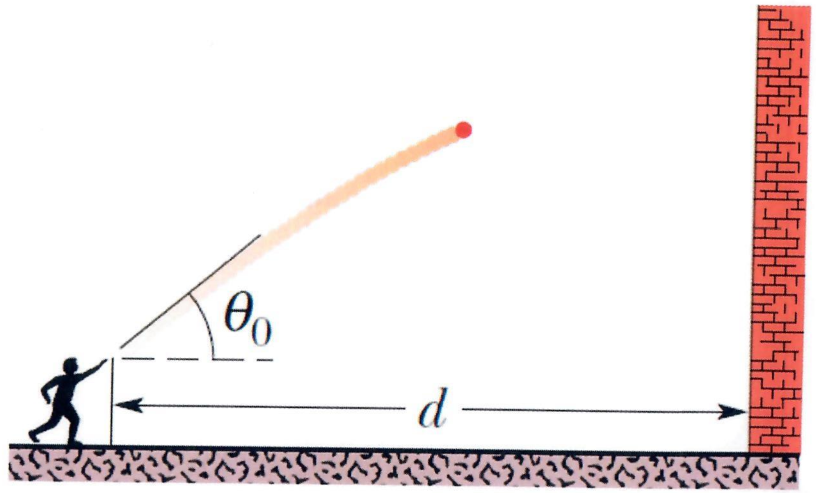
1. You throw a ball toward a wall at speed $v_o = 25.0$ m/s and at angle $\theta_o = 40.0°$ above the horizontal. The wall is distance $d = 22.0$ m from the release point of the ball. (a) How far above the release point does the ball hit the wall? (b) What are the components of its velocity as it hits the wall? (c) When it hits, has it passed the highest point on its trajectory? Give reasons for your answer.



Jy gooi 'n bal in die rigting van 'n muur teen 'n spoed $v_o = 25.0$ m/s teen 'n hoek van $\theta_o = 40.0°$ bo die horisontaal. Die muur is op 'n afstand van $d = 22.0$ m weg van die lanseerpunt van die bal. (a) Hoe ver bo die lanseerpunt sal die bal die muur tref? (b) Wat is die komponente van die bal se snelheid wanneer dit die muur tref? (c) Het die bal al die maksimum hoogte bereik toe dit die muur tref of nie? Gee redes vir jou antwoord.

(10)

Tutorial Test 3-1 — 2-D Motion

1. $d = 22.0\,m$ | $\theta_0 = 40.0°$ | $y_d = ?$

$x_0 = 0$ | $V_0 = 25.0\,m/s$ | $y_0 = 0$

$V_{0x} = V_0 \cos 40°$ | $a_y = -9.8\,m/s$

$V_{0y} = V_0 \sin 40°$

a) Time to travel to the wall:

$$\Delta x = V_{0x} t$$

$$t = \frac{\Delta x}{V_{0x}} = \frac{22.0\,m}{25.0\,m \cos 40°}$$

$$= 1.15\,s$$

$$\Delta y = (V_0 \sin\theta)t + \frac{1}{2} a_y t^2$$

$$= (25.0\,m \sin 40°)(1.15\,s) + \frac{1}{2}(-9.8\,m/s^2)(1.15\,s)^2$$

$$= 18.48\,m - 6.48\,m$$

$$= 12.0\,m$$

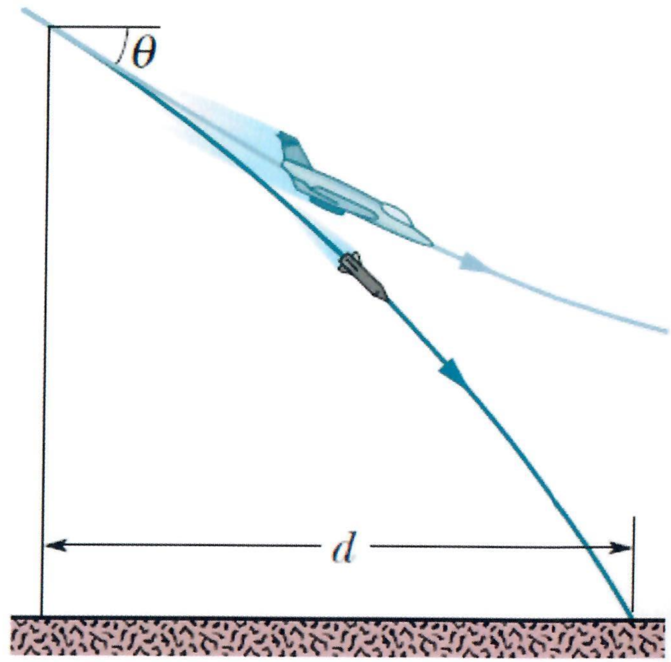
(b) $V_{0x} = V_{fx} = V_0 \cos 40° = 25.0\,m \cos 40° = 19.2\,m/s$

$V_{fy} = V_{0y} + at = 25 \sin 40 + (-9.8\,m/s^2)(1.15\,s)$

$= 4.8\,m/s$

(c) No. It has not passed the max height. The V_{yf} is a positive value, - it is still going upwards.

1. A certain airplane has a speed of 290.0 km/h and is diving at an angle of $\theta_o = 30.0°$ below the horizontal when the pilot releases a radar decoy. The horizontal distance between the release point and the point where the decoy strikes the ground is d = 700 m.
(a) How long is the decoy in the air?
(b) How high was the release point?
'n Vliegtuig het 'n spoed van 290 km/h en vlieg teen 'n hoek van $\theta_o = 30.0°$ *onder die horisontale vlak wanneer die vlieënier 'n radarstrik* (decoy) *loslaat. Die horisontale afstand tussen die loslaat posisie en waar die radarstrik land is d = 700 m.*
(a) Hoe lank is die radarstrik in die lug?
(b) Hoe hoog bo die grond is die loslaat-punt?



(10)

Tutorial Test 3-2 2-D Motion

$V_0 = 290 \text{ km/h}$ $\theta_0 = 30°$ below horizontal

$\Delta x = 700 \text{ m}$

$$V_0 = \frac{290 \text{ km}}{h} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ hr}}{60 \text{ min}} \times \frac{1 \text{ min}}{60 \text{ s}} = 80.6 \text{ m/s}$$

a) $V_{0x} = V_0 \cos\theta = 80.6 \text{ m/s} \cos 30°$

$= 69.3 \text{ m/s}$

$\Delta x = V_{0x} t$

$$t = \frac{\Delta x}{V_{0x}} = \frac{700 \text{ m}}{80.6 \text{ m/s} \cos 30°} = 10.0 \text{ s}$$

b) $V_{0y} = -V_0 \sin 30 = -80.6 \text{ m/s} \sin 30°$

$= -40.3 \text{ m/s}$

can also use $80.6 \text{ m/s} \sin(-30°)$

If ground level is zero ($y = 0 \text{ m}$)

Then

$\Delta y = V_{0y} t + \frac{1}{2} a_y t^2$

$0 - y = -40.3 \text{ m/s} (10.0 \text{ s}) + \frac{1}{2}(-9.8 \text{ m/s}^2)(10.0 \text{ s})$

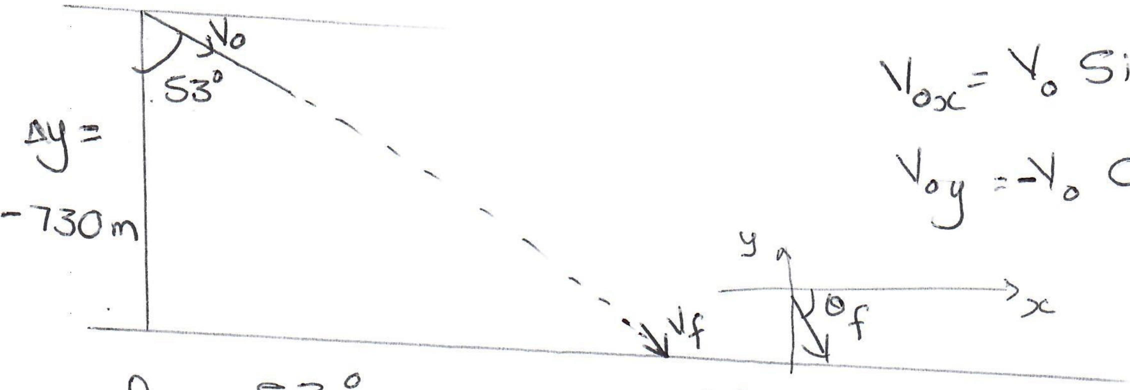
$-y = -403 \text{ m} - 490 \text{ m}$

$y = 893 \text{ m}$

1. A plane, diving with constant speed at an angle of 53.0° with the vertical, releases a projectile at an altitude of 730 m. The projectile hits the ground 5.00 s after release. (a) What is the speed of the plane? (b) How far does the projectile travel horizontally during its flight? (c) What are the horizontal and vertical components of its velocity just before striking the ground? (d) At what angle with the horizontal direction does the projectile hit the ground?

’n Vliegtuig, vlieg teen ’n konstante spoed teen ’n hoek van 53.0° met die vertikale rigting wanneer dit ’n projektiel loslaat op ’n hoogte van 730 m. Die projektiel tref die grond 5.00 s nadat dit losgelaat is. (a) Wat is die spoed van die vliegtuig? (b) Hoe ver beweeg die projektiel in die horisontale rigting tydens die vlug? (c) Bepaal die horisontale en vertikale komponente van die spoed net voor die projektiel die grond tref. (d) Wat is die hoek wat die projektiel maak met die horisontale rigting wanneer dit die grond tref? (10)

Tutorial 3-3



$V_{ox} = V_o \sin 53^\circ$

$V_{oy} = -V_o \cos 53^\circ$ (-ve downwards)

$\theta_o = 53^\circ$ $\quad$ $y_0 = 730\,m$ $\quad$ $t_f = 5.00\,s.$

$y_f = 0\,m$

a) $\Delta y = y_f - y_i$

$= V_{oy} t + \frac{1}{2} a t^2 = (-V_o \cos 53^\circ)(t) + \frac{1}{2} a t^2$

$\therefore V_o = \dfrac{\Delta y - \frac{1}{2} a t^2}{t(-\cos 53^\circ)} = \dfrac{-730\,m + \frac{1}{2}(-9.8\,m/s^2)(5\,s)^2}{(5.0\,s)(-\cos 53^\circ)}$

$= 202\,m/s$

b) $V_{ox} = V_o \sin 53^\circ \quad = 202\,m/s \sin 53^\circ$

$= 161\,m/s$

$\Delta x = V_{ox} t$

$= (161\,m/s)(5.00\,s) = 806\,m$

c) $V_{fx} = V_{ox} = 161\,m/s.$

$V_{fy} = V_{oy} + at \quad = -V_o \cos 53^\circ + \frac{1}{2}(-9.8\,m/s^2)(5.0\,s)$

$= -202\,m/s \cos 53^\circ + (-4.9)(5.0)\,m/s$

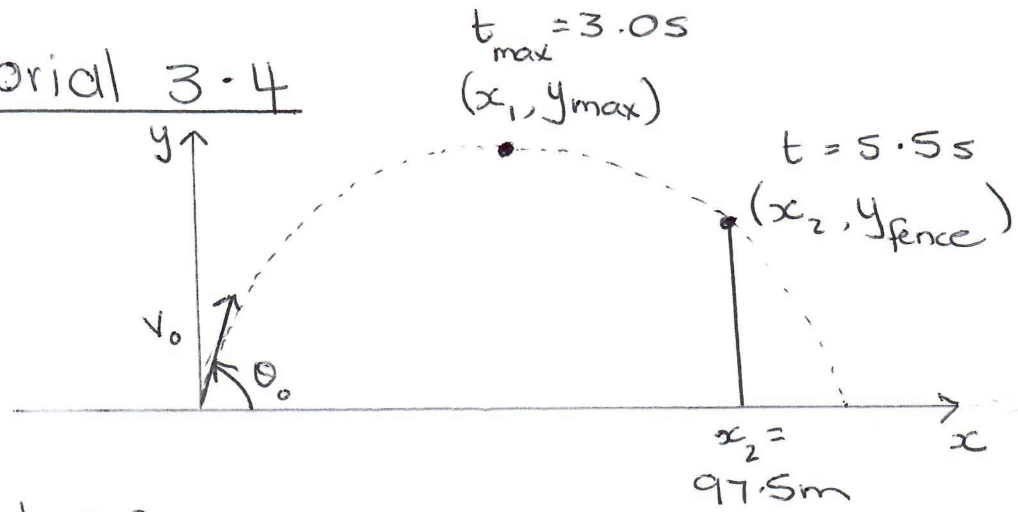
$= -171\,m/s$

d) $\theta_f = \tan^{-1}\left(\dfrac{V_{yf}}{V_{xf}}\right) = \tan^{-1}\left(\dfrac{-171\,m/s}{161\,m/s}\right)$

$= -47^\circ$ (i.e 47° below $+x$-axis)

1. A baseball is hit at ground level. The ball reaches its maximum height above ground level 3.0 s after being hit. Then 2.5 s after reaching its maximum height, the ball barely clears a fence that is 97.5 m from where it was hit. Assume the ground is level.
   (a) What maximum height above ground level is reached by the ball?
   (b) How high is the fence?
   (c) How far beyond the fence does the ball strike the ground?
   'n Bofbal word op grondvlak geslaan. Die bal bereik die maksimum hoogte bo die grondvlak 3.0 s nadat dit geslaan is. 2.5 s nadat dit die maksimum hoogte bereik het, beweeg die bal net-net oor die heining wat 97.5 m weg van die afslaan punt is. Veronderstel dat die grond gelyk is.
   (a) Wat is die maksimum hoogte van die bal bo grondvlak.
   (b) Hoe hoog is die heining?
   (c) Hoe ver anderkant die heining sal die bal die grond tref? (10)

Tutorial 3·4



At $t = 0$

$V_{oy} = V_0 \sin\theta_0$ $\qquad$ $V_{ox} = V_0 \cos\theta_0$

At $t = 3.0s$

$V_{y\,max} = 0\ m/s$ $\qquad$ $V_x = V_0 \cos\theta_0$

$y = y_{max}$ $\qquad$ $x = \frac{1}{2} x_{max} = \frac{1}{2} R$

At $t = 5.5s$

$y = y_{fence}$ $\qquad$ $x_{fence} = 97.5m$

---

a) At $t = 3.0s$ $\qquad$ $V_y = 0\ m/s$

$V_y = V_{oy} + at$

$V_{oy} = -at = -(-9.8\ m/s^2)(3\ s)$

$= 29.4\ m/s$

$\Delta y_{max} = V_{oy} t + \frac{1}{2} a t^2 = 29.4 m/s (3.0s) + \frac{1}{2}(-9.8 m/s)(3.0s)^2$

$= 44.1\ m$

Can also use $V_y^2 = V_{oy}^2 + 2a\Delta y$

b) At $t = 5.5s$

$\Delta y = V_{oy} t + \frac{1}{2} a t^2 = 29.4 m/s (5.5s) + \frac{1}{2}(-9.8 m/s^2)(5.5s)^2$

$\Delta y_{fence} = 13.5m$. Fence is just lower than 13.5 m.

Tut 3.4

c) At $t = 5.5s$ $\Delta x = 97.5m$

$$\Delta x = v_{0x}t + \frac{1}{2}a_x t^2$$
$$= v_{0x}t$$

$$v_{0x} = \Delta x / t = \frac{97.5m}{5.5s}$$
$$= 17.7 m/s.$$

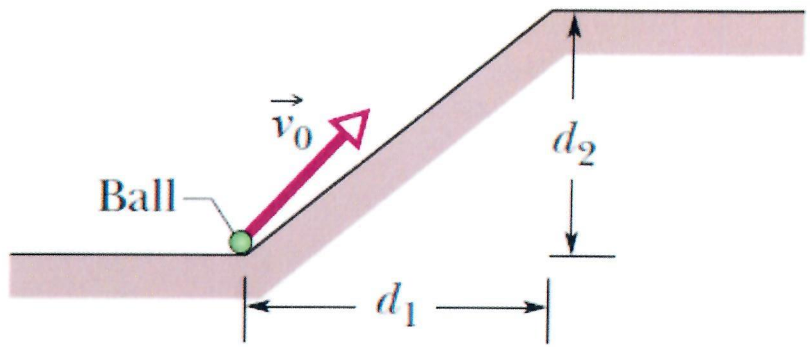
Time of flight : $t_f = 2t_{max}$
$$= 6.0s$$

$$\Delta x_{max} = R = v_{0x}t$$
$$= 17.7 m/s (6.0s)$$
$$= 106.4 m.$$

1. In the figure, a ball is launched with a velocity of magnitude 10.0 m/s, at an angle of 50.0° to the horizontal. The launch point is at the base of a ramp of horizontal length $d_1 = 6.00$ m and height $d_2 = 3.60$ m. A plateau is located at the top of the ramp.

(a) Does the ball land on the ramp or the plateau? (hint: write the height (y) as a function of the slope of the ramp)

(b) When the ball lands, what are the magnitude of its displacement from the launch point?



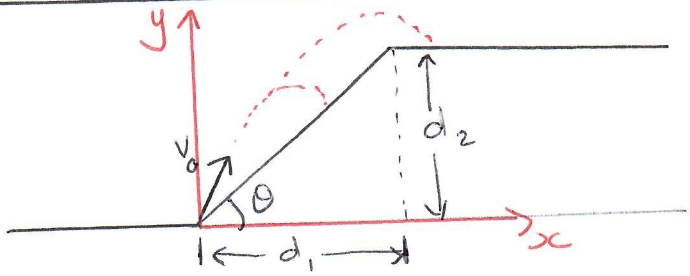
In die figuur word 'n bal gelanseer met 'n snelheid-grootte van 10 m/s teen 'n hoek van 50° ten opsigte van die horisontale rigting. Die lanseeringspunt is aan die onderpunt van 'n oprit met 'n horisontale lengte $d_1 = 6.00$ m en hoogte $d_2 = 3.60$ m. 'n Horisontale plato is aan die bokant van die oprit.

(a) Land die bal op die oprit of op die plato? (Wenk: Skryf die hoogte (y) as 'n funksie van die helling van die oprit)

(b) Wanneer die ball land, wat is die grootte van die verplasing vanaf die lanseeringspunt?

(10)

Tutorial 3-5



Place our origin at the launch point.

$V_0 = 10$ m/s $\qquad \theta = 50.0°$

$V_{0x} = V_0 \cos 50.0° = 10\text{m/s} \cos 50.0°$

$= 6.43$ m/s

To land on the plateau, $\Delta x > 6.00$m.

$\Delta x = V_{0x} t$

$t = \Delta x / V_{0x} = \dfrac{\Delta x}{V_0 \cos\theta}$ — 1

In vertical direction:

$\Delta y = (V_0 \sin\theta)t + \frac{1}{2} a_y t^2$ — 2

1 in 2 $\quad = V_0 \sin\theta \left(\dfrac{\Delta x}{V_0 \cos\theta}\right) - \frac{1}{2}(9.8\text{m/s}^2)\left(\dfrac{\Delta x}{V_0 \cos\theta}\right)^2$ 3

Slope of ramp $= \dfrac{d_2}{d_1} = \dfrac{3.60\text{m}}{6.00\text{m}} = 0.60$

$\therefore y = mx = 0.60x$ — 4

4 in 3 $\quad 0.60\cancel{x} = (\tan\theta)\cancel{x} - \dfrac{04.9x^{\cancel{2}}}{V_0^2 \cos\theta^2}$

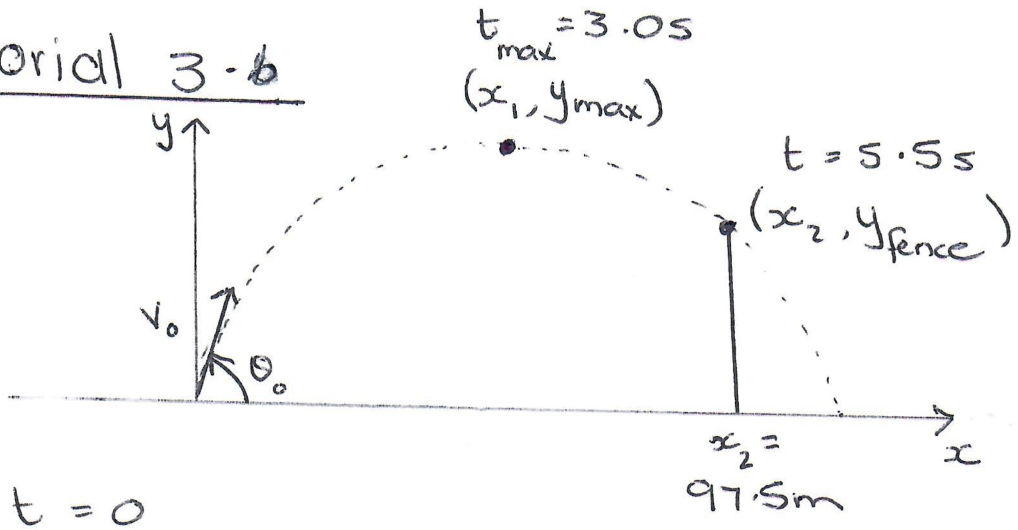
$x = 4.99$ m

This is less than 0.600 m.

The ball lands on the ramp.

1. A baseball is hit at ground level. The ball reaches its maximum height above ground level 3.0 s after being hit. Then 2.5 s after reaching its maximum height, the ball barely clears a fence that is 97.5 m from where it was hit. Assume the ground is level.
(a) What maximum height above ground level is reached by the ball?
(b) How high is the fence?
(c) How far beyond the fence does the ball strike the ground?
'n Bofbal word op grondvlak geslaan. Die bal bereik die maksimum hoogte bo die grondvlak 3.0 s nadat dit geslaan is. 2.5 s nadat dit die maksimum hoogte bereik het, beweeg die bal net-net oor die heining wat 97.5 m weg van die afslaan punt is. Veronderstel dat die grond gelyk is.
(a) Wat is die maksimum hoogte van die bal bo grondvlak.
(b) Hoe hoog is die heining?
(c) Hoe ver anderkant die heining sal die bal die grond tref? (10)

Tutorial 3.6



At $t = 0$

$V_{oy} = V_0 \sin\theta_0$ $\qquad V_{ox} = V_0 \cos\theta_0$

At $t = 3.0s$

$V_{ymax} = 0\ m/s$ $\qquad V_x = V_0 \cos\theta_0$

$y = y_{max}$ $\qquad x = \frac{1}{2} x_{max} = \frac{1}{2} R$

At $t = 5.5s$

$y = y_{fence}$ $\qquad x_{fence} = 97.5m$

---

a) At $t = 3.0s$ $\qquad V_y = 0\ m/s$

$V_y = V_{oy} + at$

$V_{oy} = -at = -(-9.8\ m/s^2(3\ s))$

$= 29.4\ m/s$

$\Delta y_{max} = V_{oy}t + \frac{1}{2}at^2 = 29.4m/s(3.0s) + \frac{1}{2}(-9.8m/s)(3.0s)^2$

$= 44.1\ m$

Can also use $V_y^2 = V_{oy}^2 + 2a\Delta y$

b) At $t = 5.5s$

$\Delta y = V_{oy}t + \frac{1}{2}at^2 = 29.4m/s(5.5s) + \frac{1}{2}(-9.8m/s^2)(5.5s)^2$

$\Delta y_{fence} = 13.5m$. Fence is just lower than 13.5 m.

Tut 3-6

c) At $t = 5.5s$ $\qquad \Delta x = 97.5m$

$$\Delta x = V_{ox}t + \frac{1}{2}a_x t^2$$
$$= V_{ox}t$$
$$V_{ox} = \Delta x / t = \frac{97.5m}{5.5s}$$
$$= 17.7 m/s.$$

Time of flight : $t_f = 2t_{max}$
$$= 6.0s$$

$$\Delta x_{max} = R = V_{ox}t$$
$$= 17.7 m/s (6.0s)$$
$$= 106.4 m.$$